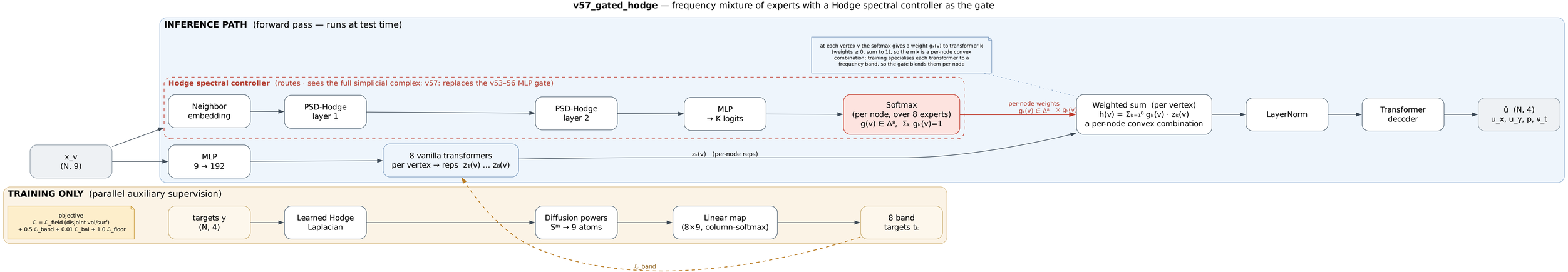
v57_gated_hodge — frequency mixture of experts with a Hodge spectral controller as the gate
INFERENCE PATH (forward pass — runs at test time)
at each vertex v the softmax gives a weight gₖ(v) to transformer k (weights ≥ 0, sum to 1), so the mix is a per-node convex combination; training specialises each transformer to a frequency band, so the gate blends them per node
Hodge spectral controller (routes · sees the full simplicial complex; v57: replaces the v53–56 MLP gate)
Neighbor embedding
PSD-Hodge layer 1
PSD-Hodge layer 2
MLP → K logits
Softmax (per node, over 8 experts) g(v) ∈ Δ⁸, Σₖ gₖ(v)=1
per-node weights gₖ(v) ∈ Δ⁸ × gₖ(v)
x_v (N, 9)
MLP 9 → 192
8 vanilla transformers per vertex → reps z₁(v) … z₈(v)
zₖ(v) (per-node reps)
Weighted sum (per vertex) h(v) = Σₖ₌₁⁸ gₖ(v) · zₖ(v) a per-node convex combination
LayerNorm
Transformer decoder
û (N, 4) u_x, u_y, p, v_t
TRAINING ONLY (parallel auxiliary supervision)
objective ℒ = ℒ_field (disjoint vol/surf) + 0.5 ℒ_band + 0.01 ℒ_bal + 1.0 ℒ_floor
targets y (N, 4)
Learned Hodge Laplacian
Diffusion powers Sᵐ → 9 atoms
Linear map (8×9, column-softmax)
8 band targets tₖ
ℒ_band